

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Replaced paper sign-in with QR-driven check-in (AWS S3); event instrumentation and data collection for 150+ scans/event, delivering real-time usage metrics for analytics and cutting processing time 80%.
- Enforced production-grade quality via GitHub Actions CI/CD with 90% coverage gates across 20+ tests for testing and validation.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Led cross-functional team of 6; introduced Neo4j to offload recursive queries from PostgreSQL, cutting p95 48% (850ms → 440ms) and enabling multi-hop data analysis and analytics.
- Refactored Cypher with relationship-type indexing; traversals 3× faster than SQL self-joins, removed full-table scans.
- Implemented GitLab CI/CD (build, test, containerized deploy) with branch protection, 85% coverage across 35+ tests for requirements validation.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery* May 2025 – Nov. 2025

- Engineered data pipeline and multi-client voice assistant that batches natural dialogue, resolves implicit speaker intent, and dispatches real tasks across concurrent voice streams.
- Reduced API response time 73% by offloading RAG pipeline for semantic data analysis to async Redis/Celery; deployed Python FastAPI on AWS Lambda (cloud), cutting cold-start 86% (6,643ms → 930ms); key metrics and KPIs for pipeline health.
- Built Electron + Next.js client with embedded HTTP server for OAuth and real-time IPC/webhook events (sub-50ms propagation).

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow* Feb. 2025

- Won Interactive Media Track: Gesture-controlled wizard duel where full-body movements cast spells, block attacks, and trigger abilities against an AI opponent.
- Built full-body gesture pipeline in Python using MediaPipe Pose to map 33 skeletal landmarks into game actions; optimized inference loop to sustain 60 FPS with sub-20ms input lag.
- Developed custom 2D vector physics engine with AABB collision detection to accurately translate live 3D skeletal coordinates to game state, maintaining <5px drift at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash
Frameworks Spring Boot, FastAPI, Node.js, Angular, React, Next.js, Redis
Databases PostgreSQL, Neo4j, MongoDB
Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitLab CI/CD, CI/CD Pipelines
Tools & Data Grafana, Excel, Microsoft Office, data analysis, telematics